

Landslide and Debris Flow Survival Guide

This guide covers warning signs, immediate survival actions, and post-landslide safety for landslides, mudflows, and debris flows. Based on USGS, FEMA, Red Cross, and NDMA guidelines. India is highly vulnerable — the Western Ghats, northeastern states, Uttarakhand, and Himachal Pradesh experience frequent deadly landslides, especially during monsoon season.

SPEED WARNING: Debris flows can move at 50 to 100 kilometres per hour — faster than you can run. If you are in a high-risk area during or after heavy rainfall, do not wait to see the slide. Leave immediately when you see the warning signs.

SECTION 1: WARNING SIGNS — RECOGNISE THESE IMMEDIATELY

A landslide may be imminent if you see or hear any of the following:

Sounds: A faint rumbling sound that grows louder. Trees cracking. Boulders knocking together. A roaring sound like a rushing river where no river exists.

Ground changes: New cracks appearing in the ground, road surface, or building foundations. Ground bulging at the base of a slope. Soil moving away from the base of walls or trees. Fence lines or walls that are suddenly out of alignment.

Building changes: Doors or windows that suddenly stick or jam for the first time. New cracks in plaster, tile, brick, or concrete floors. Stairs or floors that tilt or crack.

Water signs: New springs or water seeping from slopes that were previously dry. A stream that suddenly goes from clear to muddy — this indicates debris flow activity upstream. A sudden decrease in stream water level even though rain is still falling.

KEY RULE: Many landslide deaths occur at night when people are asleep during heavy rainfall. If you live in a landslide-prone area and heavy rain has been falling for several hours, stay awake and alert during the night.

SECTION 2: IMMEDIATE SURVIVAL ACTIONS

If you are outdoors:

Step 1. Run immediately to the nearest high ground that is out of the path of the landslide. Move uphill and away from the slide direction — do not run parallel to the flow.

Step 2. If you cannot escape, move to the side of the slide path. Landslides tend to follow valleys and drainage channels — moving sideways off the path may save your life.

Step 3. If escape is completely impossible, curl into a tight ball, protect your head with your arms, and try to hold onto a tree or large fixed object.

NEVER cross a road, bridge, or stream where water or mud is flowing. Debris flows can travel under bridges and wash them away instantly.

If you are indoors:

Step 1. Move to the highest floor of the building and to the downhill side of the structure (the side facing away from the slope the landslide is coming from).

Step 2. Open downhill-facing windows and doors. This allows debris to pass through rather than building up pressure and collapsing the structure.

Step 3. Do NOT open uphill-facing windows or doors — this lets the flow into the building.

Step 4. Get on TOP of sturdy furniture (table, bed) rather than sheltering behind it. You want to be as high as possible above the debris flow, not trapped underneath furniture.

Step 5. Protect your head with your arms and hold on to something fixed.

DO NOT go to the basement or lowest floor. Debris flows fill lower areas first. Moving down is moving toward the danger.

SECTION 3: IF YOU ARE TRAPPED AFTER A LANDSLIDE

Stay calm and conserve energy.

Signal for rescuers by tapping on walls, pipes, or any hard surface — three taps, pause, repeat. Use a whistle if available.

Send an SMS with your location if you have any phone signal.

Cover your mouth and nose with cloth to avoid inhaling mud and debris dust.

Do not eat or drink muddy or contaminated water — even if you are very thirsty. See the water purification guide for safe water.

Stay as still as possible. Moving around risks destabilising the debris and causing further collapse.

SECTION 4: AFTER THE LANDSLIDE — IMMEDIATE SAFETY

Step 1. Stay well away from the slide area. Additional slides are very common immediately after the first one — the slope is already destabilised.

Step 2. Do not enter damaged buildings. Check for structural damage before approaching — cracks in load-bearing walls, a sagging roof, or a leaning structure mean the building may collapse.

Step 3. Watch for flooding. Landslides often dam streams and rivers. When the natural dam breaks, a sudden flash flood can follow. Move away from stream channels.

Step 4. Check for injured people near (but not inside) the slide area. Direct professional rescuers to their location — do not attempt to enter the debris field yourself.

Step 5. Report broken utility lines — gas, electricity, water — to authorities immediately. Do not attempt to repair them yourself.

Step 6. Do not use tap water until authorities confirm it is safe. Landslides contaminate water supplies with soil, sewage, and chemicals.

WHEN TO RETURN HOME: Do not return to your home until a professional geotechnical assessment has confirmed it is safe. The slope that failed once can fail again, especially during subsequent rainfall.